

Mason Mitchell

(424) 413-6200 | mmitchell7@tulane.edu | masonmitchell.dev | Los Angeles, CA

EDUCATION

Tulane University

Bachelor of Science in Computer Science & Information Technology

New Orleans, LA

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Systems & Networks, Linear Algebra, Data Science, Discrete Mathematics

EXPERIENCE

Tulane University - Biomedical Engineering

Undergraduate Researcher

New Orleans, LA

Mar 2026 - Present

- Developing segmentation workflows for hematopoietic cell analysis using Python, YOLOv8, Roboflow, and segmentation frameworks (YOLOv26, SAM, MedSAM).
- Building image preprocessing and data-augmentation pipelines to standardize data and support model training and evaluation.
- Collaborating with biomedical engineering researchers to translate research needs into reproducible software and machine-learning workflows.

Albert Lepage Center for Entrepreneurship - Tulane University

AI Tech Fellow

New Orleans, LA

Sep 2026 - Present

- Selected for the 2026-27 Lepage Tech Fellows cohort to support student founders exploring software and applied-AI solutions.
- Provide peer-to-peer technical consulting and pitch preparation for community-partner projects through the Lepage Center.

The Hidden Genius Project

Senior Youth Educator & International Program Ambassador

Inglewood, CA

Mar 2022 - Present

- Directed computer science workshops for 100+ students, teaching Python, programming logic, and full-stack development through project-based instruction.
- Represented the organization at London Black Tech Fest 2024 and mentored students through technical projects and problem-solving exercises.

PROJECTS

Proxify LLC | Python, TypeScript, Tailscale, 3proxy

- Architected a remotely managed proxy infrastructure platform combining custom Python APIs, TypeScript services, Tailscale networking, and 3proxy for distributed node deployment.
- Built automated provisioning and secure remote-administration workflows to reduce manual configuration across Linux hosts.

ResearchBridge: A Platform for Researchers | TypeScript, PostgreSQL (Supabase), REST APIs

- Architecting a platform that matches students with faculty research opportunities based on academic interests, technical skills, and available openings.
- Designing database, authentication, permissions, search/filter, and resume-scanning features for an end-to-end student-to-professor experience.

Greenlab: Computer Vision Cell-Analysis Platform | TypeScript, Python, YOLOv8, Roboflow, Supabase

- Built a full-stack computer vision platform for hematopoietic analysis, integrating model inference, dataset management, authentication, and application data.
- Developed preprocessing and augmentation pipelines and extended the system toward segmentation-based cell analysis.

TECHNICAL SKILLS

Languages: Python, C/C++, Swift, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, SwiftUI, Node.js

AI / Computer Vision: YOLOv8, Image Segmentation, Image Preprocessing, Data Augmentation, Roboflow, Jupyter

Tools / Backend: Linux/UNIX, Git/GitHub, Supabase, Firebase, Railway, Cloudflare, Clerk, Stripe, Tailscale, 3proxy